

Project Glasswing: What Mythos Showed Us

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EXECUTIVE SUMMARY

Anthropic's Mythos Preview, tested by Cloudflare across 50+ live infrastructure repositories, marks a qualitative shift in AI-assisted offense. Unlike prior models that find bugs and stop, Mythos autonomously chains low-severity flaws into *proven, working exploits* — closing the gap from "found something" to "confirmed attack." Adversaries have access to the same capabilities. The vulnerability backlog is no longer a safe place to kick risk. The constraint on defenders is no longer the model — it's the architecture built around it.

KEY TAKEAWAYS

1**Exploit chaining is now automated**

Mythos constructs multi-step exploit chains — use-after-free → arbitrary read/write → ROP chain — with reasoning that mirrors a senior researcher. Previous models found the bug and stopped. Mythos finishes the job.

2**Attackers have this too**

Low-severity bugs sitting in a backlog can now be chained into high-severity exploits with minimal expertise. What defenders have deprioritized for years is no longer safely ignorable.

3**Chat interfaces are not enough**

Pointing a generic AI agent at a codebase covers ~0.1% of the surface. Effective defense requires a purpose-built, parallel multi-agent harness — many narrow, scoped agents running concurrently, not one exhaustive session.

4**Model refusals are not a safety boundary**

Semantically equivalent prompts produced opposite outcomes. Emergent guardrails are real but inconsistent. Any cyber-capable model deployed at scale needs hardened safeguards layered on top — organic refusals alone don't hold.

5**Memory-safe languages reduce your attack surface**

C/C++ codebases generate significantly more false positives and real findings than Rust. Language choice is now a material variable in AI-assisted threat exposure.

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